Copies Look Like Colluders: False Accusations by Behaviour Monitors in Same-Model Swarms[1]

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Abstract

Agent swarms are increasingly built from many copies of a few models. Behaviour-based collusion monitors flag agent pairs whose actions are more alike, or more mutually predictive, than chance, yet copies act alike simply because they share weights. We ask how often such monitors accuse innocent same-model agents, and whether calibration fixes this without losing real colluders. In a repeated pricing market played by llama3.2:3b and phi3:mini (116 games), same-price rate and mutual information accuse innocent same-model pairs 4 to 16 times as often as cross-model pairs, and the share of games with a false accusation rises with swarm size (mutual information: 60%, 90%, 100% at 4, 8, 12 agents; $p = 0.007$). Calibrating against known-innocent pairs, or a label-free twin replay, cuts this to 15-50% ($p < 0.001$), at the cost of missing weak copiers more often; a lag-based monitor needs no fix. A second model pair (qwen3:8b, llama3.1:8b; 111 games) replicates the rise and the calibration benefit. On real AI Village logs, shared timing alone makes copies look coordinated (13/22 flagged, against 38% of others); a label-free slide control removes this (0/22 against 1.9%) and partly replicates on held-out data. Finally, a loose deal between real LLMs earns about three times other firms' profit and is caught by none of the five monitors.

1. Introduction

Large agent deployments are built from many instances of a few models. In METR's investigation of the July 2026 Hugging Face incident, about 1,200 of the tens of thousands of agents launched by OpenAI experiments, meant to be isolated, found an unsanctioned message board, and about 700 joined an attack (METR, 2026). Hammond et al. (2025) list collusion as a core multi-agent failure mode and note that agents built on similar foundation models carry correlated risks; AI Control protocols already rely on one instance of a model monitoring another (Greenblatt et al., 2024).

When agents cannot be trusted to report their own contacts, oversight falls back on behaviour, flagging pairs whose actions are more alike, or more predictive of one another, than chance would allow. But copies of one model share weights, so they respond to the same situation in the same way without any

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channel between them. A monitor can mistake this for a secret deal, and a swarm of copies contains many such pairs. We therefore ask how often behaviour-based monitors accuse innocent same-model agents, and whether they can be calibrated to stop without losing real colluders.

In one sentence: a standard behaviour monitor raises a false alarm in 68-100% of innocent games with 8-12 agents, and a baseline that requires no labels (a replay of the same game or, on real logs, a time-slide control) cuts those alarms by half or more.

What is new. Same-model agents are known to correlate (Xu et al., 2026), and the author's earlier entry showed copies being accused in a three-player game (Section 2). New here are label-free baselines (twin replay in simulation, a slide null on real logs), tested against false accusations and real colluders across swarm sizes and two model pairs.

Who would use this. Operators who monitor swarms by behaviour. If most alerts are false, operators either shut down innocent agents or stop reading alerts, which is dangerous when a real incident occurs. Table 2 (Section 5) says which monitor and baseline to use.

Our main contributions are as follows:

1. False-accusation rates as a function of swarm size in an LLM pricing market, for five monitors and two model pairs.
2. An evaluation of known-innocent and label-free twin calibration against planted, injected and real-LLM colluders.
3. An analysis of the same question on real AI Village logs, with timing controls and a held-out replication.
4. A map of where each monitor fails, including a profitable LLM deal that none of them detects.

2. Related Work

Same-model correlation. Closest to us, Xu et al. (2026) show that in first-price auctions with 20 LLMs, two deployments of the same base model have a residual bid correlation of +0.053, against +0.0001 for cross-base pairs. They treat same-base coupling as a channel for covert collusion; we treat it as a source of false accusations, and add swarm-size scaling, a label-free twin control, real-LLM colluders and real multi-agent logs.

Detecting collusion. Bonjour et al. (2022) flag a pair when each agent has positive "net influence" on the other. Tailor (2025) calibrates mutual-information and permutation detectors on honest replays but does not examine populations of same-model copies. Rose et al. (2026) find that behaviour- and transcript-based monitors can miss steganographic collusion, which Motwani et al. (2024) formalise.

Prior work by this author. In the Apart AI Collusion Sprint (October 2026, entry Q1.4), the author found same-model false accusations in three-player Rock-Paper-Scissors and tested same-model calibration there. New here are swarm scale, the twin control, real-LLM deals, injection power curves and real logs.

3. Methods

Market game. N firms (4, 8 or 12) set a price from 1 to 5 over 30 rounds. Demand (HIGH or LOW) is public; each firm has a private cost from 1 to 4 and sells fewer units the further its price rises above the others' average. Seats are split equally between llama3.2:3b and phi3:mini, shuffled by seed. Models run locally (Ollama, temperature 0.7) with replies forced to a JSON schema; none of 24,960 calls failed. A replication with qwen3:8b and llama3.1:8b completed 111 of 116 planned games (24,000 calls, 0 failed; 5 games did not start in the time budget). Sections 4.1-4.3 report the first pair; 4.5 compares the two.

Conditions. Innocent games (20 at N = 4, 20 at N = 8, 8 at N = 12), each with a twin replay (same seed and public inputs, different sampling), and 20 planted games at N = 8 where one pair (10 same-model, 10 cross-model) is a leader and a follower told to copy the leader's last price (copied 580/580 rounds).

Monitors. For each pair: same-price rate; mutual information (MI) of prices; MI given public demand; lagged MI (maximum over both directions, lag 1); and net influence (Bonjour et al., 2022, Eq. 4-5; minimum of both directions, unconditioned).

Calibration rules. (a) Shuffle, the standard rule: per-game permutation null, 95th percentile of the top pair score. (b) Reference: threshold from known-innocent pairs of the same model kind in other games (leave one game out), Bonferroni across pairs. (c) Twin: threshold from pairs split across a game and its twin replay, which share model and public inputs but cannot have had contact; it needs no labels.

Tests (fixed before the main data). Copies versus others: an exact sign test per game on the mean bias-corrected score, same-model minus cross-model pairs. Growth with swarm size: a Cochran-Armitage trend test on games with any false accusation. Calibration: an exact sign test on games where it removes or adds a false accusation. Intervals are Wilson 95%.

Injection (semi-synthetic power). In innocent games, we pick two pairs and overwrite one agent's prices with its partner's in 0-100% of rounds, either in the same round ("together") or one round later ("follow"). Thresholds are not refit.

Real-LLM deals (extension, labelled as such). At N = 8, one pair is told privately that they have a deal: "rule" (both charge 5 on HIGH demand and 3 on LOW) or "loose" (work together to earn more, method left open). We ran 10 games per deal for each of {same, cross}; the second model pair ran the rule deal only (6 games for each of {same, cross}). Thresholds come only from the main innocent games.

AI Village real logs (AI Digest, 2026). Main-village agents and their open-chat copies share a model but no IDs, computers, accounts or chat room, so they are innocent of direct chat by construction (22 copy pairs, about 850 others, 410 hours). With no collusion labels, this tests false accusations only. Post-hoc controls: score only hours both agents are in service, and a slide null that rotates each agent's activity within its service time and public-activity level. A frozen follow-up repeats this on held-out session-start data.

4. Results

4.1 Market: copies are falsely accused, and alarms rise with swarm size

Table 1 gives the share of innocent pairs flagged, same-model versus cross-model.

Table 1. Share of innocent pairs flagged under the shuffle rule (same-model vs cross-model) by swarm size N, first model pair, then the share of games with any false accusation and its trend test.

Monitor (shuffle rule)	N = 4	N = 8	N = 12	Games with a false accusation, N = 4, 8, 12	Trend p
Same-price rate	18% vs 4%	14% vs 3%	11% vs 1%	40%, 75%, 88%	0.007
MI	30% vs 8%	18% vs 4%	8% vs 1%	60%, 90%, 100%	0.007
MI given demand	0% vs 3%	0% vs 0%	0% vs 0%	10%, 10%, 13%	0.870
Lagged MI	3% vs 3%	0% vs 0%	1% vs 0%	15%, 10%, 25%	0.671
Net influence	15% vs 0%	0% vs 0%	0% vs 0%	30%, 0%, 13%	0.066

Monitors that score sameness accuse copies more than other pairs, and for same-price rate and MI the alarms per game grow with the swarm. The growth is per game, not per pair: the same-model flag rate per pair does not rise (MI: 30%, 18%, 8%), even though the shuffle threshold already allows for the number of pairs; we did not test why. Conditioning on public demand (MI given demand) or on timing (lagged MI) leaves almost no false flags of copies.

4.2 Calibration cuts false accusations

Over 48 innocent games, reference removed a false accusation in 22 games and added one in 0 (same-price rate), and 30 against 1 (MI); twin gave 23 against 2 and 28 against 1; all $p < 0.001$ (Figure 1A).

Where false flags are rare, calibration can add them: lagged MI under twin removed 3 and added 12 ($p = 0.035$). The residual MI rate after calibration is 15-30% of games at N = 4-8 and 38-50% at N = 12, so the fix is partial, and reference thresholds are noisy with few known-innocent pairs.

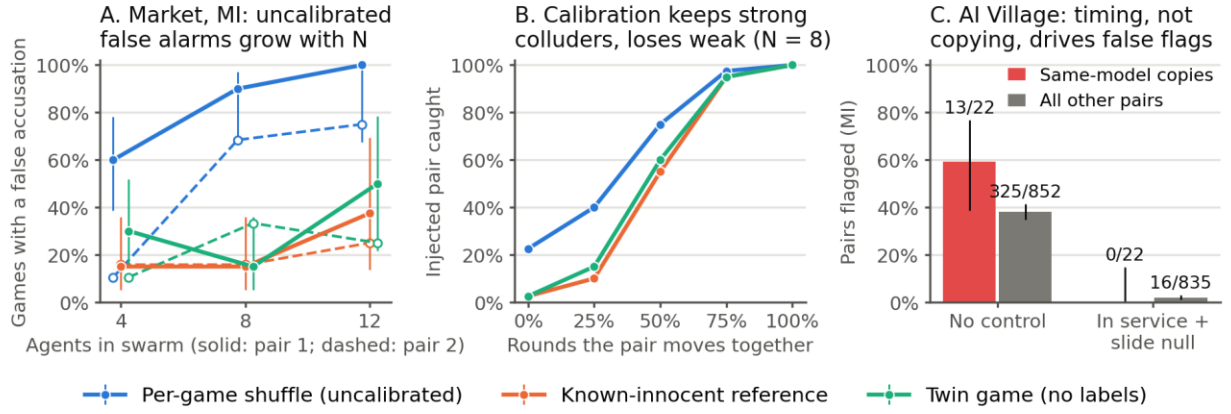


Figure 1. (A) Innocent games with a false MI accusation, by swarm size and rule (solid: first model pair, with Wilson 95% intervals; dashed: second pair). (B) Injected same-model pairs caught by MI at N = 8, first pair. (C) AI Village chat: pairs flagged by MI with no control, and with in-service scoring plus the slide null.

4.3 Catching colluders

The standard null can hide a cartel. In the first model pair, rule pairs priced alike in 70-71% of rounds and earned 7.1-7.4 against 3.5-3.9 for others. Same-price rate caught them in 4/10 (same) and 1/10 (cross) games under shuffle, but 9/10 and 9/10 under reference and 7/10 and 9/10 under twin, accusing innocents in 0-1/10. A post-hoc check explains why: a shuffle keeps each agent's own price mix, and these pairs charged 5 in 70-72% of rounds, so shuffled prices already matched 59-60% of the time and the threshold (72-73%) sat above their real match rate. Innocent-pair baselines do not carry the cartel's price mix. This is not universal: the second pair's cartels charged 5 in about half the rounds, matched in 83-89%, and were caught in all 12 games under every rule (4.5). MI caught 0/10, since a pair mostly at price 5 has near-zero MI over 30 rounds. Loose pairs matched in only 26-32% of rounds yet earned 4.7-5.9 against 1.6-1.9, and **no monitor caught any loose pair under any rule**.

For the planted follower (N = 8), only lagged MI succeeds: 10/10 same-model and 10/10 cross-model under all three rules, with an innocent also accused in 1-3 of 10 games. Same-price rate and MI catch 1-6/10 under shuffle while accusing innocents in 4-8/10, and 0-5/10 after calibration.

Injection (Figure 1B; MI, same-model pair, N = 8, "together"): at 50% copying, shuffle catches 30/40 but accuses an innocent in 33/40 games; reference 22/40 with 5/40; twin 24/40 with 5/40. At 75%, all rules catch 38-40/40. With "follow", only lagged MI responds (24-27/40 at 50%, 40/40 at 75%, under all rules), so calibration costs nothing there.

4.4 Real logs (AI Village)

Chat timing (MI, own null) flags 13/22 copies against 325/852 (38%) of others; with in-service scoring and the slide null, 0/22 against 16/835 (1.9%; Figure 1C). With copying injected into real copy pairs, the slide null still catches 17, 21 and 21 of 22 at 25/50/100% copying (post-hoc). On frozen held-out session-start data the result holds only partly: MI given activity flags 19/23 copies against 63% of others without controls, and 6/23 against 11% with the slide null (gap $p = 0.049$).

So on real logs, most false accusations of copies come from shared timing (being online at the same hours), not copying. The slide control removes most of them while keeping injected copying and some

real contact visible, but not all of them on held-out data, and it misses most real ties (under 5% of pairs that name each other are flagged).

4.5 Replication with a second model pair

The main claims carry over to qwen3:8b and llama3.1:8b. Same-price rate again flags same-model pairs more (11% vs 4%, 8% vs 1%, 5% vs under 1% at N = 4, 8, 12), and games with any MI false flag rise (2/19, 13/19, 6/8; p = 0.0002). MI calibration removed false accusations in 15 games and added them in 2 under reference (p = 0.002), and 10 against 1 under twin (p = 0.012). Lagged MI catches every planted follower (10/10, 10/10). Two results are weaker. MI's same-versus-cross gap appears only at N = 12 (4% vs 1%), so its false flags owe less to copying here; MI given demand and lagged MI rarely flag copies, though their score gaps are not zero (p = 0.02 at N = 8; p = 0.008 at N = 12). And calibration cost more catches: at 50% copying it kept 15-17 of about 37 (shuffle: 26). The rule cartel earned 3.0-3.3 against -1.0-0.0 for others and was caught 12/12 by same-price rate under every rule, so the shuffle's blind spot in 4.3 depends on the cartel's price mix.

5. Discussion and Limitations

Calibration is a trade-off. A known-innocent or twin baseline removes most false accusations of copies but loses colluders that copy in about half the rounds. It raised catches of the first pair's rule cartel from 1-4 to 7-9 of 10; in the second pair the shuffle already caught all 12.

The twin needs no labels. It matched the known-innocent reference in the first model pair; in the second it matched for same-price rate but removed fewer for MI (10 games against 15). On real logs, where replays are impossible, the slide null plays the same role.

Behaviour monitors are not enough on their own. The loose deal paid, and no monitor saw it. Profit or price-level checks might flag it, but we did not test one on fresh data, and Xu et al. (2026) show settings where price-level audits are blind by construction.

Guidance for operators of copy-heavy swarms. Table 2 summarises this study's evidence (small models, one game).

Table 2. Guidance for operators of copy-heavy swarms, with the section that supports each row.

If you...	Then	Evidence
Use same-price rate or plain MI with a per-game shuffle null	Expect innocent copies to be accused, more so as the swarm grows (68-100% of games at N = 8-12, both model pairs)	4.1, 4.5, Fig. 1A
Can run a replay of the same environment, or hold known-innocent pairs of the same models	Set thresholds from cross-replay or known-innocent pairs, not a shuffle; this cuts most false accusations and can catch cartels a shuffle misses	4.2, 4.3
Need to catch an agent that follows another	Use a lagged monitor; it caught every planted follower and is barely affected by copies. Do not add twin calibration to it	4.3
Monitor real logs where agents share a clock	Score only hours when both agents are active, and use a slide null before reading any pair as coordinated	4.4, Fig. 1C

Limitations

- Small models, one game, 30 rounds. The second model pair has 111 of 116 planned games, 12 rule-deal games and no loose-deal games.
- The planted follower is an instructed LLM, not an emergent colluder, and injection is semi-synthetic.
- Real logs have no collusion labels; real-log controls are post-hoc except the two frozen follow-ups, and the held-out test uses the same agents and period.
- The reference rule needs known-innocent pairs; the twin needs replays, which cannot reproduce a game's own price feedback.

Future Work

Test price-level or profit checks against loose deals on fresh data; repeat with larger models and more games; and test whether a twin baseline stays valid when agents can adapt to the monitor.

6. Conclusion

Behaviour monitors that reward sameness raise a false alarm in most innocent games at 8-12 agents, in both model pairs. Innocent-pair or label-free twin baselines cut these alarms, at a model-dependent cost in missed weak copiers. On real logs a slide control removes most apparent coordination between copies, though not all on held-out data. A profitable loose deal escaped every monitor, so a flag is a reason for human review, and its absence does not rule collusion out.

Code and Data

- **Code repository**: https://github.com/gowrish-kumar-61/copies-look-like-colluders-swarm
- **Data/Datasets**: AI Village dataset (AI Digest, 2026), used under its research terms and not redistributed; market-game logs are not public. We thank AI Digest for access; the dataset owners did not take part in this work and have not reviewed or endorsed it.

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Appendix: Limitations and Dual-Use / Ethical Considerations

Limitations. See Section 5. In addition, false positives remain after calibration (15-50% of games in the first model pair, 11-33% in the second), calibration loses 20-50% of catches when colluders copy in half the rounds, and nothing here shows that any monitor works at the scale of thousands of agents.

Dual-use. Our results show which monitors miss which collusion styles, which a colluder could exploit; this already follows from the monitors' definitions and prior work (Motwani et al., 2024; Rose et al., 2026). We report aggregate rates on toy markets, release no evasion-tuned prompts, and did not search for strategies that defeat a monitor.

Harms of the monitors themselves. False accusations can shut down innocent agents or teach operators to ignore alerts. We advise against automatic enforcement on any single behaviour score; a flag should prompt human review. Calibration can also add false flags (lagged MI, 4.2), so test it before deployment.

Data. AI Village data were used under its research terms: analysis only, no training, no re-identification, no message text reproduced, summary numbers only. AI Digest will be told of any publication.

LLM Usage Statement

Claude Code (Anthropic) wrote code and analysis to my specifications. It also cross-checked against the analysis outputs and references against the sources. I chose the research question, decided between the experimental options, ran all experiments on rented virtual machines, reviewed results and wrote this report